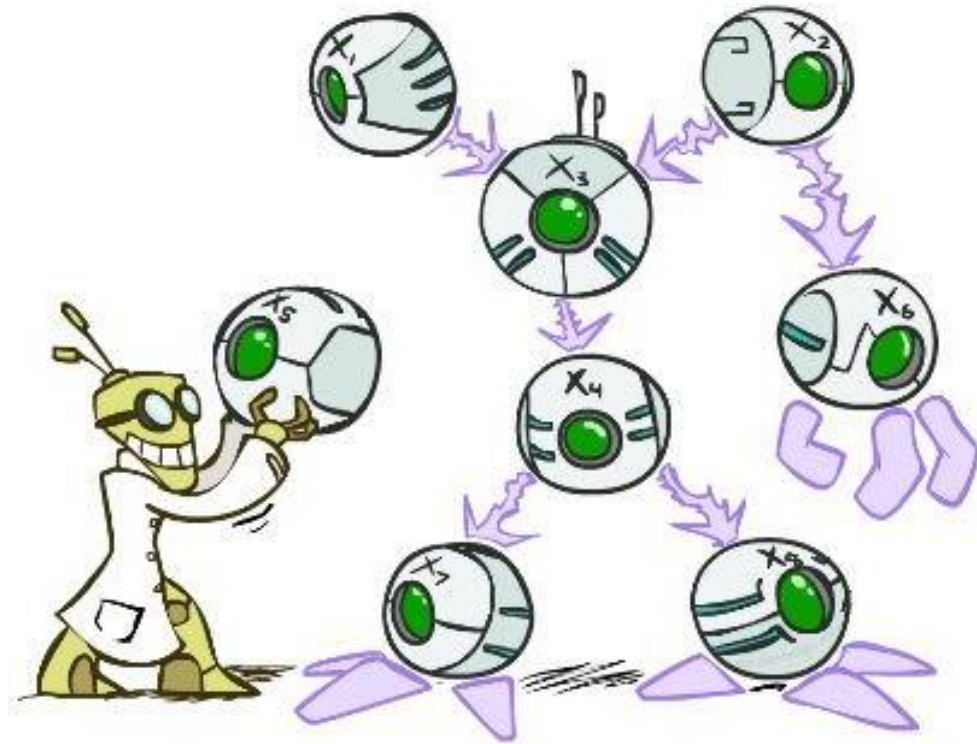


Artificial Intelligence

Bayesian Networks

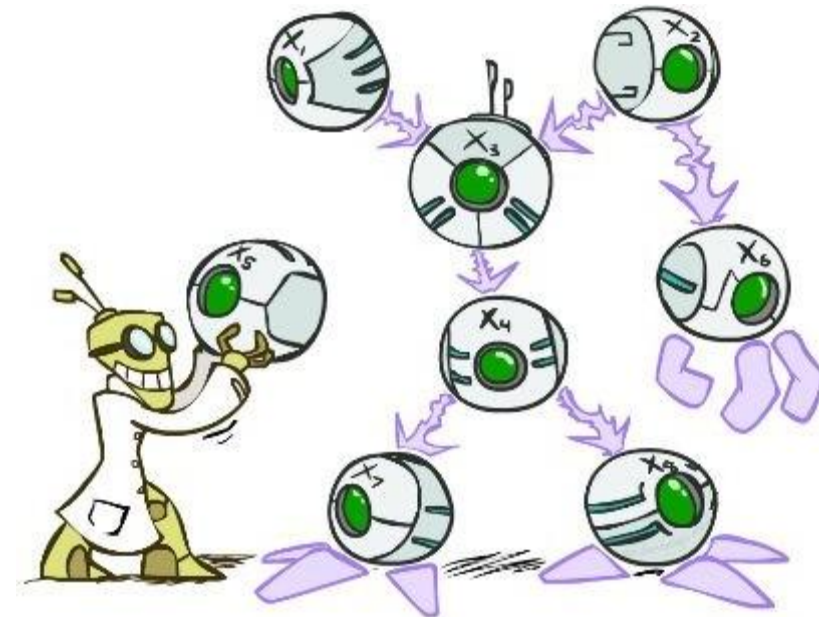


Reminder: elementary probability

- Basic laws: $0 \leq P(\omega) \leq 1$ $\sum_{\omega \in \Omega} P(\omega) = 1$
- Events: subsets of Ω : $P(A) = \sum_{\omega \in A} P(\omega)$
- Random variable $X(\omega)$ has a value in each ω
 - Distribution $P(X)$ gives probability for each possible value x
 - Joint distribution $P(X, Y)$ gives total probability for each combination x, y
- Summing out/marginalization: $P(X=x) = \sum_y P(X=x, Y=y)$
- Conditional probability: $P(X|Y) = P(X, Y)/P(Y)$
- Product rule: $P(X|Y)P(Y) = P(X, Y) = P(Y|X)P(X)$
 - Generalize to chain rule: $P(X_1, \dots, X_n) = \prod_i P(X_i | X_1, \dots, X_{i-1})$

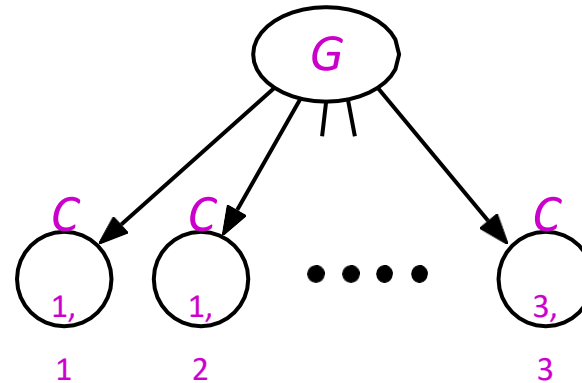
Bayes Nets: Big Picture

- **Bayes nets:** a technique for describing complex joint distributions (models) using simple, conditional distributions
 - A subset of the general class of graphical models
- Use local causality/conditional independence:
 - the world is composed of many variables,
 - each interacting locally with a few others
- Outline
 - **Representation: syntax and semantics**
 - Exact inference
 - Approximate inference



Graphical Model Syntax

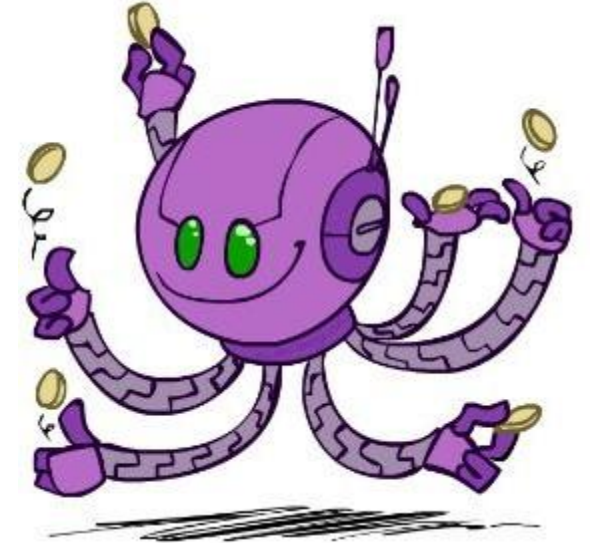
- **Nodes: variables (with domains)**
 - Can be assigned (observed) or unassigned (unobserved)
- **Arcs: interactions**
 - Indicate “direct influence” between variables
 - Formally: absence of arc encodes conditional independence (more later)



0.11		0.11
0.11	0.11	0.11
0.11	0.11	0.11

Example: Coin Flips

- n independent coin flips

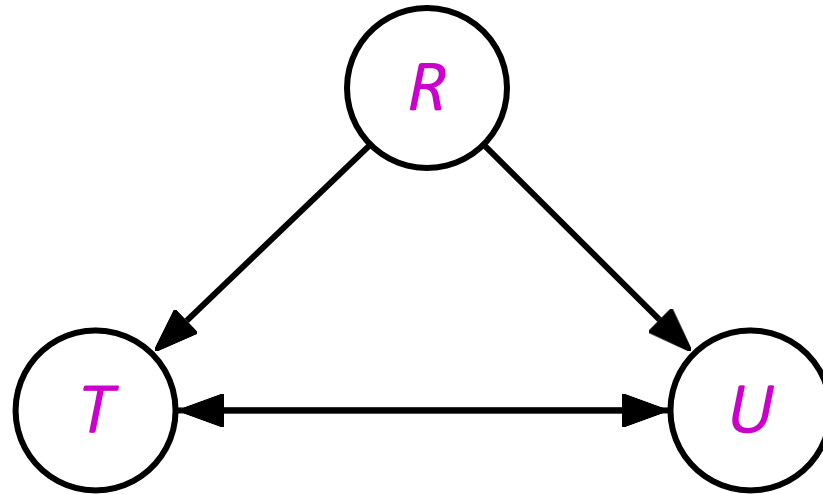


- No interactions between variables: absolute independence

Example: Traffic

- Variables:

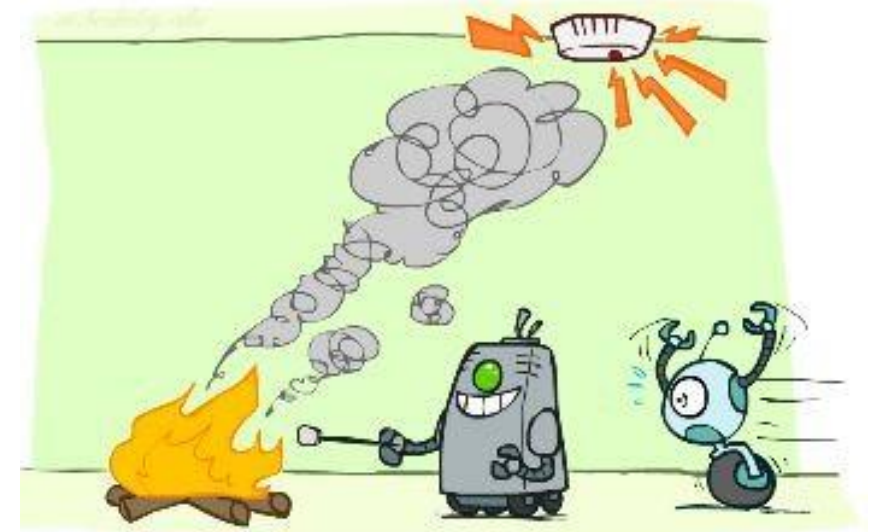
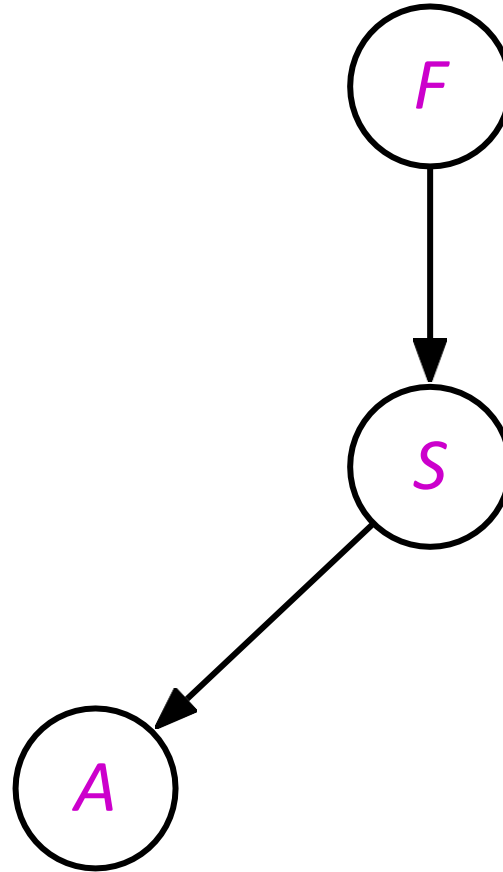
- T: There is traffic
- U: I'm holding my umbrella
- R: It rains



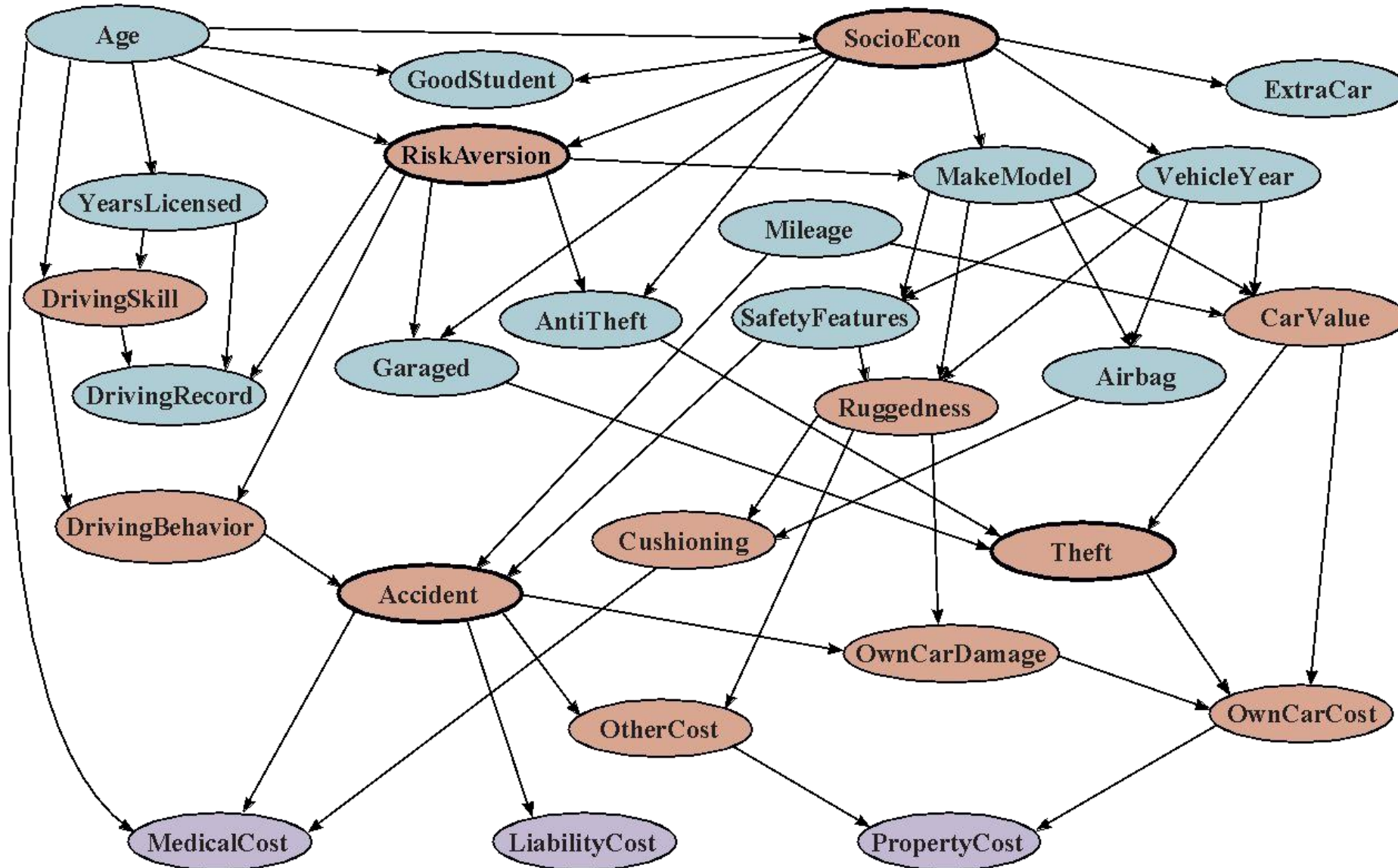
Example: Smoke alarm

- Variables:

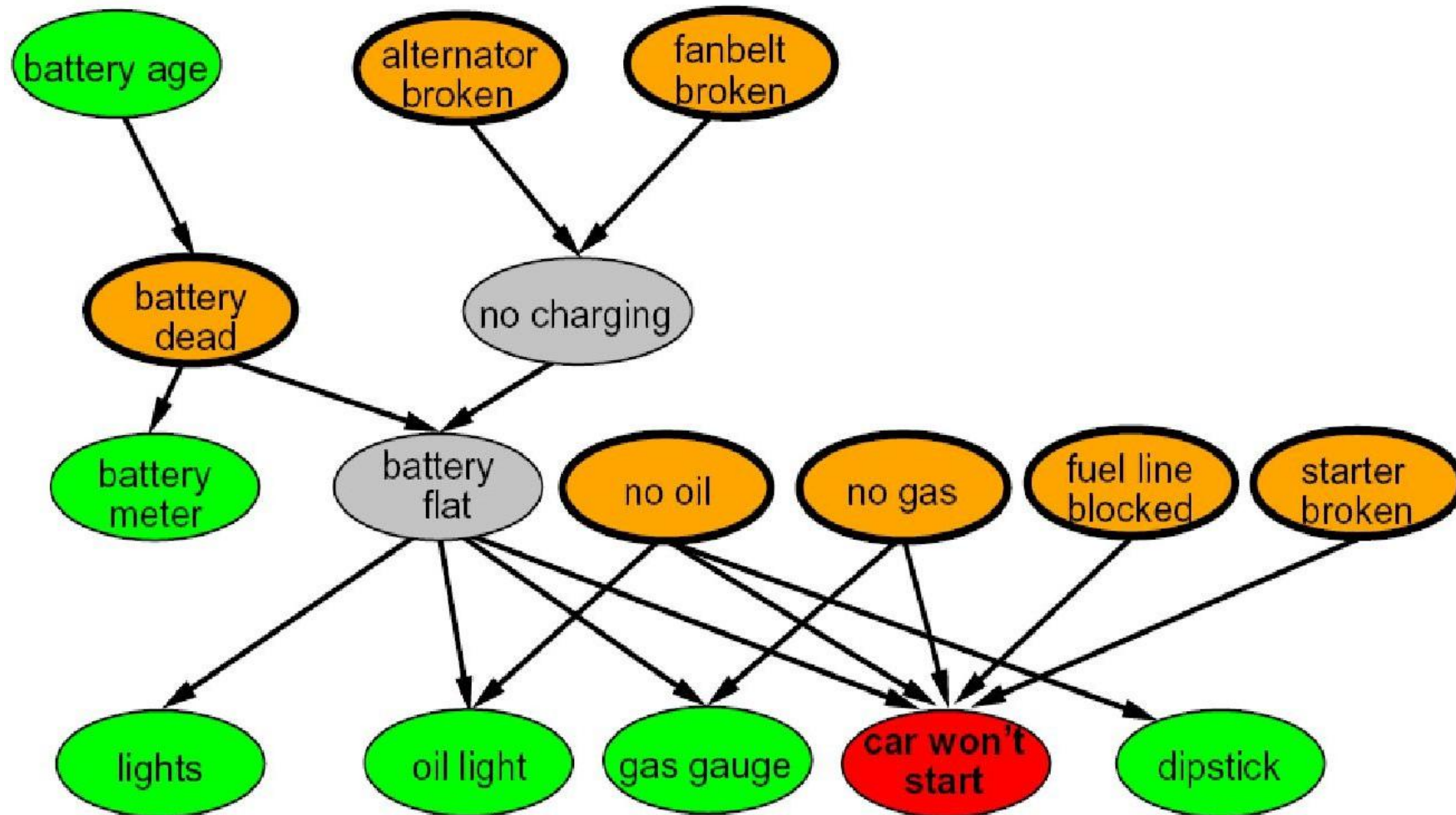
- F: There is fire
- S: There is smoke
- A: Alarm sounds



Example Bayes' Net: Car Insurance



Example Bayes' Net: Car Won't Start



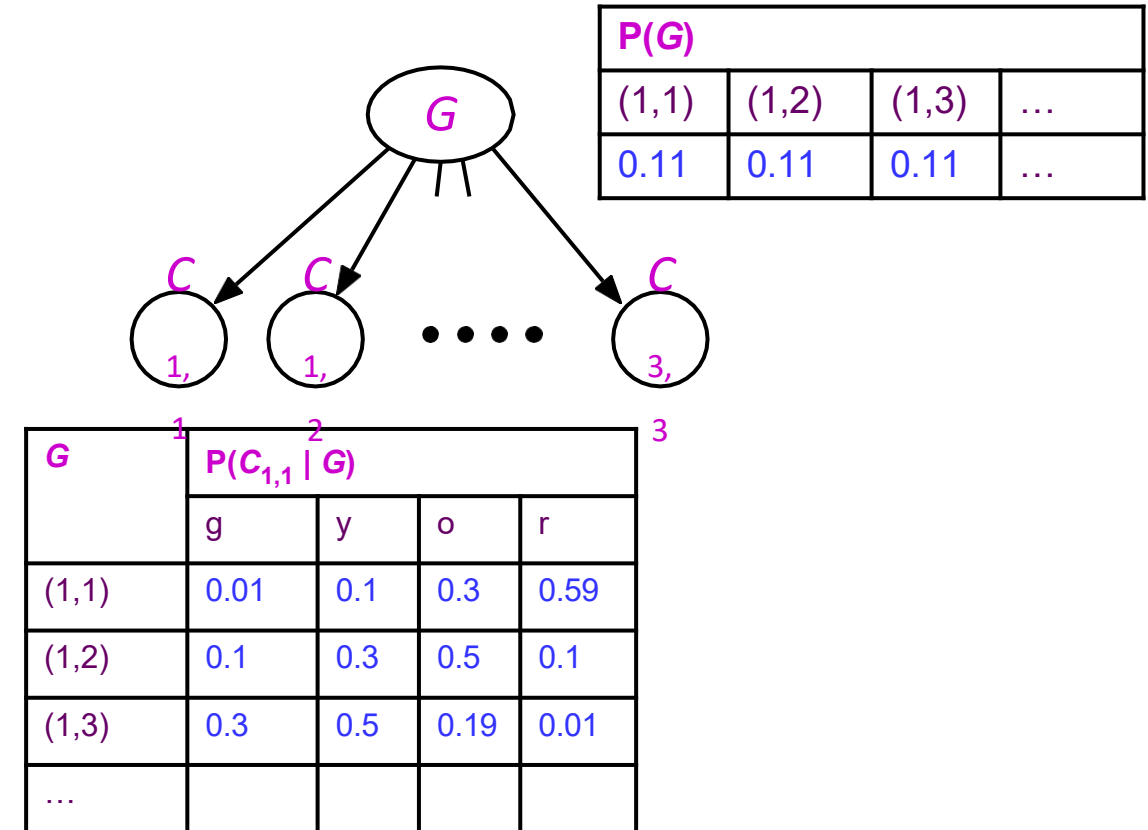
Bayes Net Syntax and Semantics



Bayes Net Syntax

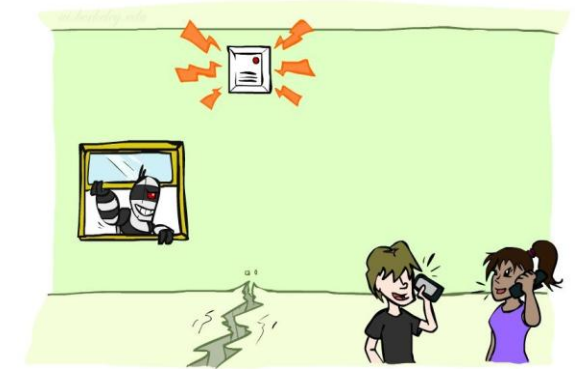
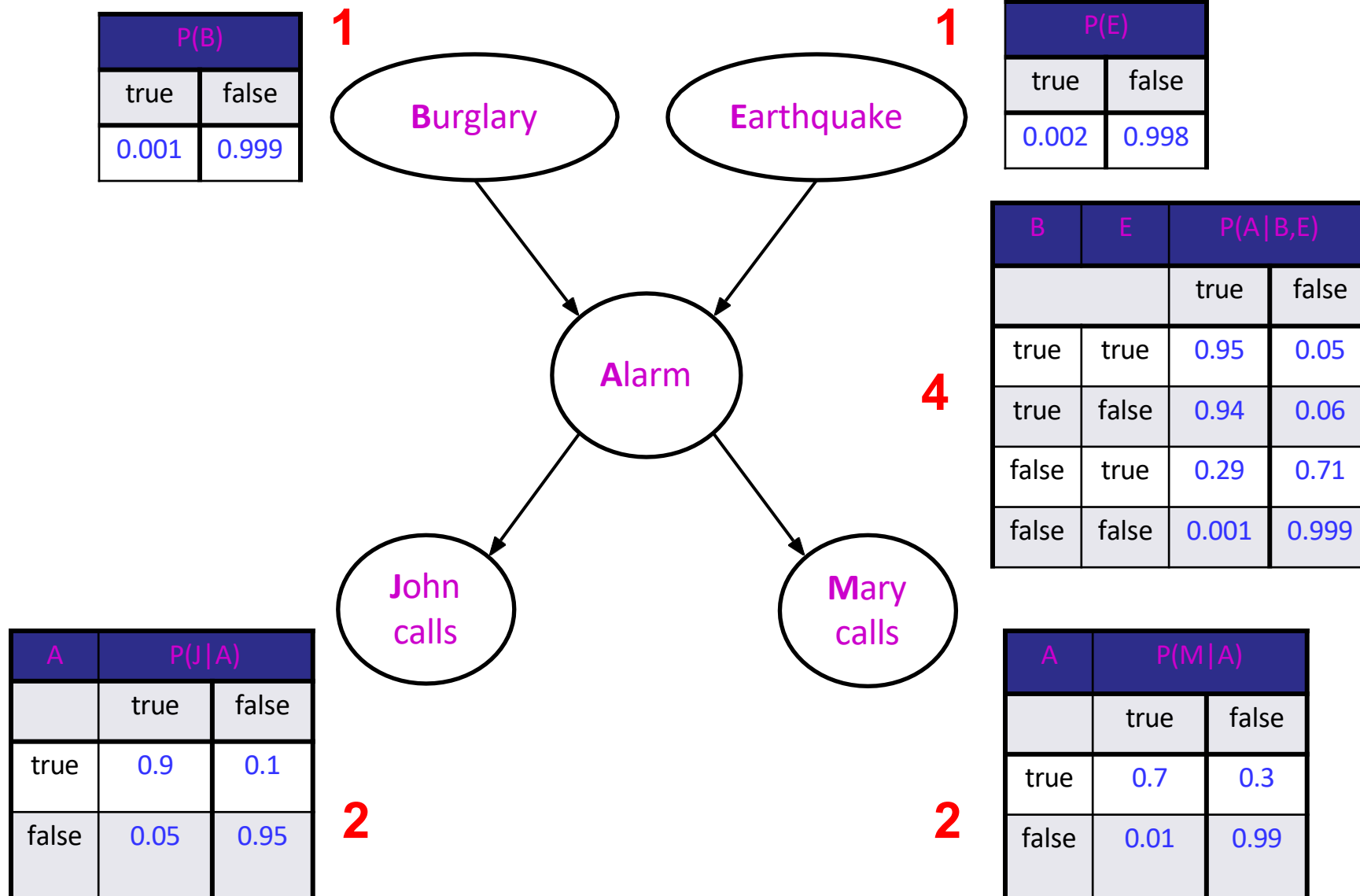


- A set of nodes, one per variable X_i
- A directed, acyclic graph
- A conditional distribution for each node given its **parent variables** in the graph
 - **CPT** (conditional probability table); each row is a distribution for child given values of its parents



Bayes net = Topology (graph) + Local Conditional Probabilities

Example: Alarm Network



Number of *free parameters* in each CPT:

Parent range sizes $d_1, \dots, d_k$

Child range size d

Each table row must sum to 1

$$(d-1) \prod d_i$$

General formula for sparse BNs

- Suppose
 - n variables
 - Maximum range size is d
 - Maximum number of parents is k
- Full joint distribution has size $O(d^n)$
- Bayes net has size $O(n \cdot d^k)$
 - Linear scaling with n as long as causal structure is local

Bayes net global semantics

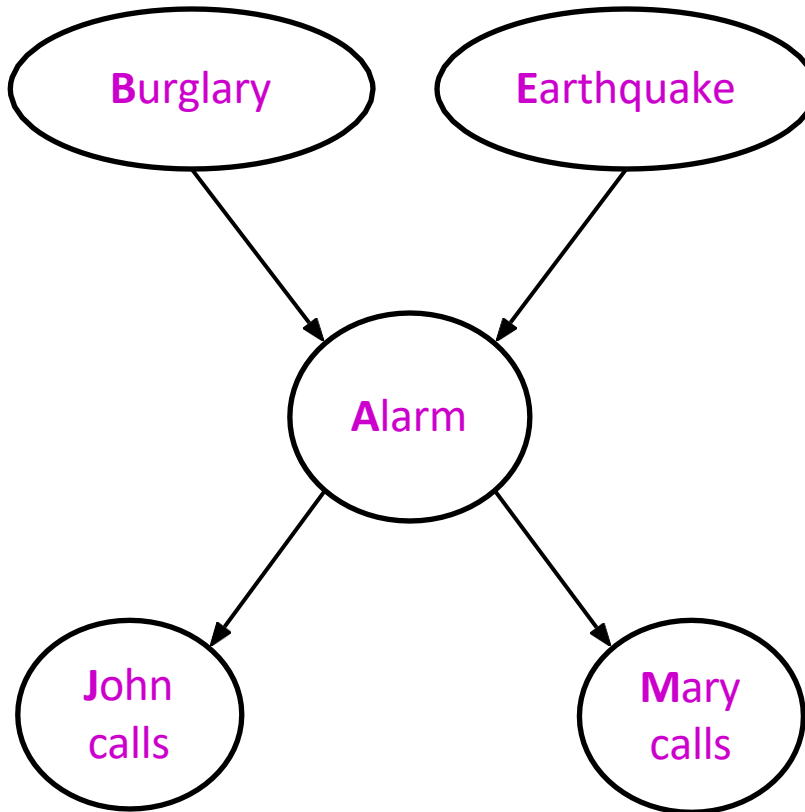


- Bayes nets encode joint distributions as product of conditional distributions on each variable:

$$P(X_1, \dots, X_n) = \prod_i P(X_i \mid \text{Parents}(X_i))$$

Example

P(B)	
true	false
0.001	0.999



P(E)	
true	false
0.002	0.998

$$P(b, \neg e, a, \neg j, \neg m) =$$

$$P(b) P(\neg e) P(a|b, \neg e) P(\neg j|a) P(\neg m|a)$$

$$=.001 \times .998 \times .94 \times .1 \times .3 = .000028$$

B	E	P(A B,E)	
		true	false
true	true	0.95	0.05
true	false	0.94	0.06
false	true	0.29	0.71
false	false	0.001	0.999

A	P(J A)	
	true	false
true	0.9	0.1
false	0.05	0.95

A	P(M A)	
	true	false
true	0.7	0.3
false	0.01	0.99

Conditional independence in BNs



- Compare the Bayes net global semantics

$$P(X_1, \dots, X_n) = \prod_i P(X_i \mid \text{Parents}(X_i))$$

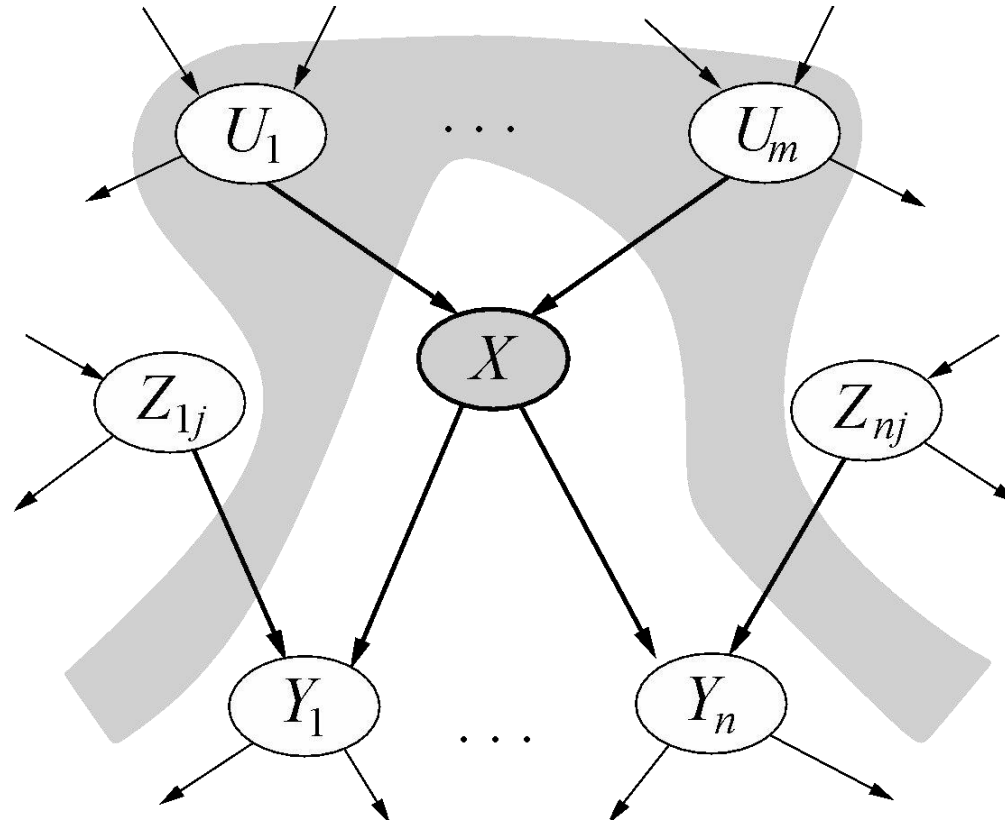
with the chain rule identity

$$P(X_1, \dots, X_n) = \prod_i P(X_i \mid X_1, \dots, X_{i-1})$$

- Assume (without loss of generality) that $X_1, \dots, X_n$ sorted in topological order according to the graph (i.e., parents before children), so $\text{Parents}(X_i) \subseteq X_1, \dots, X_{i-1}$
- So the Bayes net asserts conditional independences $P(X_i \mid X_1, \dots, X_{i-1}) = P(X_i \mid \text{Parents}(X_i))$
 - To ensure these are valid, choose parents for node X_i that “shield” it from other predecessors

Conditional independence semantics

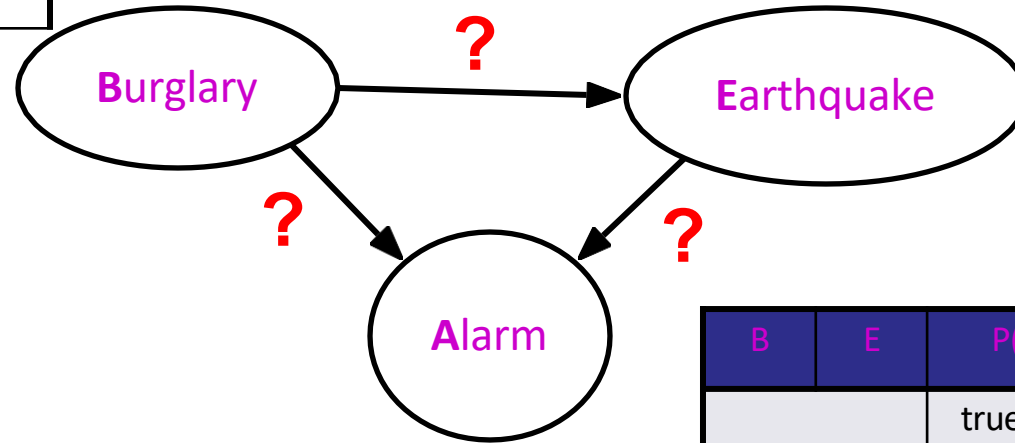
- *Every variable is conditionally independent of its non-descendants given its parents*
- Conditional independence semantics $\Leftrightarrow$ global semantics



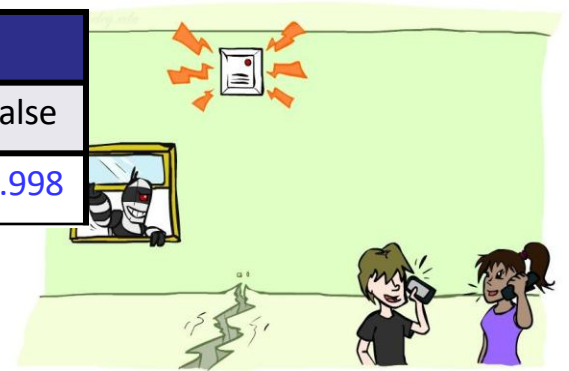
Example: Burglary

- Burglary
- Earthquake
- Alarm

P(B)	
true	false
0.001	0.999



P(E)	
true	false
0.002	0.998

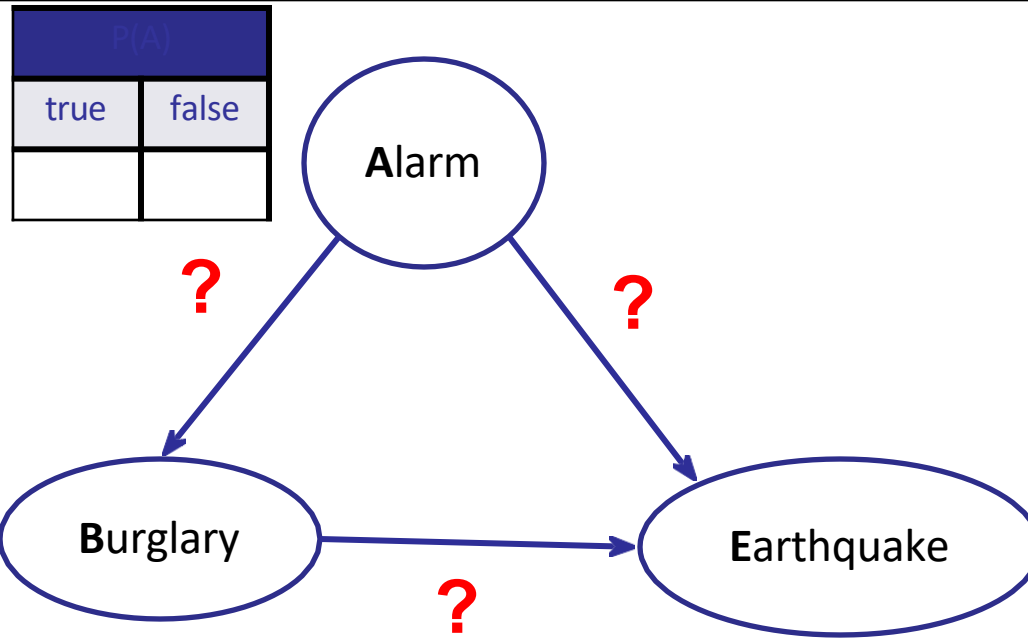


B	E	P(A B,E)	
		true	false
true	true	0.95	0.05
true	false	0.94	0.06
false	true	0.29	0.71
false	false	0.001	0.999

Example: Burglary

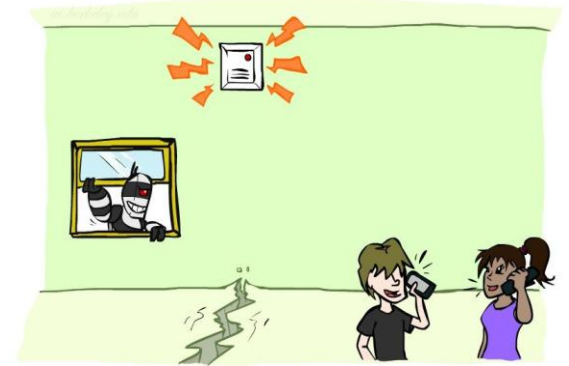
- Alarm
- Burglary
- Earthquake

A	P(B A)	
	true	false
true	?	
false		



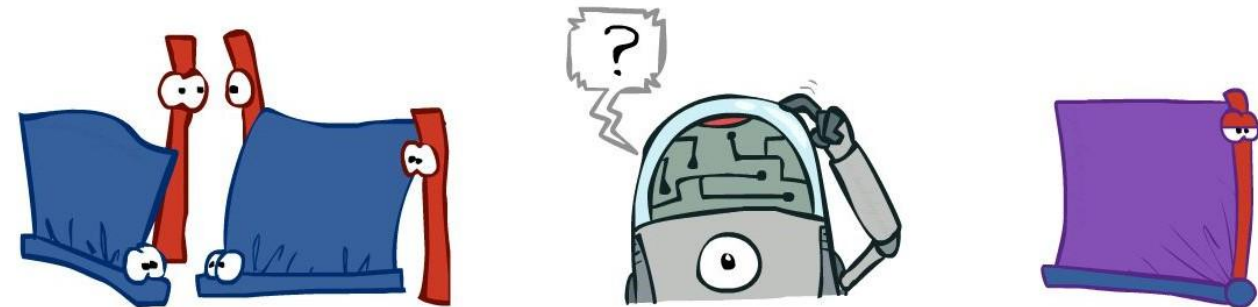
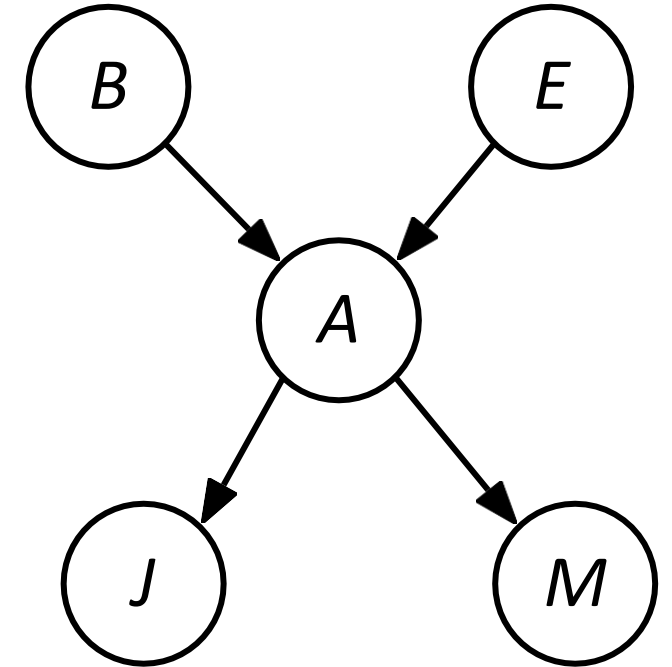
P(A)	
true	false

A	B	P(E A,B)	
		true	false
true	true	?	
true	false		
false	true		
false	false		



Inference by Enumeration in Bayes Nets

- Reminder of inference by enumeration:
 - Any probability of interest can be computed by summing entries from the joint distribution: $P(Q \mid e) = \alpha \sum_h P(Q, h, e)$
 - Entries from the joint distribution can be obtained from a BN by multiplying the corresponding conditional probabilities
- $P(B \mid j, m) = \alpha \sum_{e,a} P(B, e, a, j, m)$
- $= \alpha \sum_{e,a} P(B) P(e) P(a \mid B, e) P(j \mid a) P(m \mid a)$
- So inference in Bayes nets means computing sums of products of numbers: sounds easy!!
- Problem: sums of **exponentially many** products!



Can we do better?

- Consider $uwy + uwz + uxy + uxz + vwy + vwz + vxy + vxz$
 - 16 multiplies, 7 adds
 - Lots of repeated subexpressions!
- Rewrite as $(u+v)(w+x)(y+z)$
 - 2 multiplies, 3 adds
- $\sum_{e,a} P(B) P(e) P(a|B,e) P(j|a) P(m|a)$
- $= P(B)P(e)P(a|B,e)P(j|a)P(m|a) + P(B)P(\neg e)P(a|B,\neg e)P(j|a)P(m|a)$
 $+ P(B)P(e)P(\neg a|B,e)P(j|\neg a)P(m|\neg a) + P(B)P(\neg e)P(\neg a|B,\neg e)P(j|\neg a)P(m|\neg a)$
Lots of repeated subexpressions!

Summary

- Independence and conditional independence are important forms of probabilistic knowledge
- Bayes net encode joint distributions efficiently by taking advantage of conditional independence
 - Global joint probability = product of local conditionals
- Exact inference = sums of products of conditional probabilities from the network

